Software for management of knowledge organization systems

https://doi.org/10.5281/zenodo.15672523

Jakob Voß[1]

[1]Verbundzentrale des GBV (VZG)

2026-11-12

This report is much more usable in HTML at https://bartoc.org/software!

The terminology registry **BARTOC** collects information about all kinds of **knowledge organization systems** (KOS) such as controlled vocabularies, classifications, ontologies, and other semantic artifacts. This report summarizes information about software for KOS management.

Scope

This report is limited to **specialized KOS software for editing, vizualizing and/or analyzing terminologies.** The term "terminology" in this document is used as synonym for all kinds of KOS. Software is excluded if:

- it is not aimed at KOS management in particular but more generic software such as database management systems, content management systems and text editors,
- it is more specific software, restricted to a limited type of KOS, such software for linguistic terminology management and identifier systems,
- it cannot be used as stand-alone application, like programming libraries, and hosted services.

Software for **ontology alignment** is only included if it is more then a research prototype and if we managed to get it running. See Bergman (2018) for a comprehensive list of software in this category. The Ontology Alignment Evaluation Initiative (OAEI) motivates the creation of tools for automated ontology alignment.

KOS software

This report includes **74** instances of KOS software so far. The list is managed in a JSON file as array of CodeMeta records with fields as described below. Applications without license or repository are closed source. Deprecated applications are included for historical reference.

Table 1

name	system	edit	api	languages	license	repository	update	category
Semantic MediaWiki	Web			PHP	GPL	github	2026	SoftwareImage, editor
Wikibase	Web			PHP	GPL	wikimedia	2026	ServerApplication, database, editor
TemaTres	Web			PHP	GPL	github	2026	ServerApplication, editor
iQvoc	Web			Ruby	Apache	github	2026	SoftwareImage, WebApplication, editor
VocBench	Web			Java, JavaScript	BSD	bitbucket	2026	WebApplication, editor
Skosmos	Web	-		PHP	MIT	github	2026	WebApplication, viewer
Cocoda	Web		-	JavaScript	MIT	github	2026	WebApplication, mapping editor
Terminologue	Web		-	JavaScript	MIT	github	2026	WebApplication, editor
SSSOM Curator			-	Phython	MIT	github	2026	WebApplication, CommandLineApplication, mapping editor
JSKOS Server	Web			JavaScript	MIT	github	2026	ServerApplication, WebAPI, database
OpenTheso	Web			Java	GPL	github	2026	SoftwareImage, WebApplication, ServerApplication, WebAPI
SkoHub Vocabs	Web			JavaScript	Apache	github	2026	WebApplication, SoftwareImage, viewer
Ontology Lookup Service (OLS)	Web	-		JavaScript, Java	Apache	github	2026	ServerApplication, WebAPI, WebApplication, SoftwareImage
ABECTO	JVM	-	-	Java	Apache	github	2026	CommandLineApplication, SoftwareImage, validator
voc4cat-tool	CLI, GitHub, Excel	-	-	Python	BSD	github	2026	CommandLineApplication, converter
Protégé	JVM		-	Java	BSD	github	2026	Application, editor
OntoPortal Alliance	Web	-		Ruby	BSD	github	2026	ServerApplication, WebAPI, WebApplication, SoftwareImage
ShowVoc	Web	-	-	JavaScript	BSD	bitbucket	2026	WebApplication, viewer
VocExcel	Excel, CLI	-	-	Python	BSD	github	2026	CommandLineApplication, converter
jskos-cli	CLI	-	-	JavaScript	MIT	github	2026	CommandLineApplication, converter, validator
sssom-py	CLI	-	-	Python	MIT	github	2026	CommandLineApplication, converter
TS4NFDI API Gateway	Web	-		Java, JavaScript	MIT	github	2026	WebAPI, ePackage, viewer, converter
jskos-proxy	Web	-		JavaScript	MIT	github	2026	WebAPI, SoftwareImage, viewer, converter
Atramhasis	Web			Python	GPL	github	2026	WebApplication, editor
sssom-java	CLI	-	-	Java	GPL	github	2026	CommandLineApplication, converter
sssom-js	CLI	-	-	JavaScript	MIT	github	2026	CommandLineApplication, converter
Snowstorm	Web	-		Java	Apache	github	2026	WebAPI, SoftwareImage, erverApplication, viewer
Network of Terms	Web	-		Typescript	EUPL	github	2026	WebAPI, SoftwareImage, converter
LODE	Web	-		Java	ISC	github	2026	SoftwareImage, erverApplication, viewer
WIDOCO	JVM	-	-	JavaScript, Java	Apache	github	2026	CommandLineApplication, SoftwareImage, viewer
OxO	Web	-		JavaScript	Apache	github	2026	WebApplication, WebAPI, SoftwareImage, erverApplication,
WebVOWL	Web	-	-	JavaScript	MIT	github	2026	WebApplication, SoftwareImage, viewer
O'FAIRe	Web	-		Java	MIT	github	2026	WebApplication, WebAPI, erverApplication, validator
OnToology	Web, git	-	-	Python	Apache	github	2026	WebApplication, viewer, validator
FOOPS!	Web	-	-	Java, JavaScript	Apache	github	2026	WebApplication, CommandLineApplication, validator
TerminoloGit	Web, git	-	-	Python	GPL	gitlab	2026	WebApplication, erverApplication, converter
Onto4ALL Editor	Web		-	JavaScript, PHP	Apache	github	2026	WebApplication, erverApplication, editor
Ontology Development Kit	CLI	-	-	Python	BSD	github	2026	CommandLineApplication, SoftwareImage, converter, validat

Table 1

name	system	edit	api	languages	license	repository	update	category
Themis	Web	-		Java	Apache	github	2026	WebApplication, WebAPI, SoftwareImage, CommandLineAp
Vocabseditor	Web			Python	MIT	github	2025	WebApplication, WebAPI, SoftwareImage, erverApplication,
MSCR				Java	MIT	github	2025	WebApplication, WebAPI, mapping editor
Chowlk	Web	-	-	Python	Apache	github	2025	WebApplication, WebAPI, SoftwareImage, CommandLineAp
SkoHub Shapes	RDF	-	-	SHACL	?	github	2025	SoftwareImage, validator
EVOKS	Web		-	Python	MIT	github	2025	SoftwareImage, CommandLineApplication, editor, converter
OOPS!	Web	-	-	Java	?	github	2025	WebApplication, CommandLineApplication, viewer
VocPub Profile	RDF	-	-	SHACL	CC	github	2025	eLibrary, validator
ODM2 Controlled Vocabularies	Web			Python	BSD	github	2025	WebApplication, WebAPI, editor
qSKOS	CLI	-		Java	GPL	github	2024	CommandLineApplication, validator
Jekyll RDF	Web	-		Ruby	MIT	github	2024	CommandLineApplication, SoftwareImage, viewer
VocPrez	Web	-		Python	GPL	github	2024	WebApplication, WebAPI, SoftwareImage, viewer
Amalgame	Web		-	prolog	BSD	github	2023	WebApplication, SoftwareImage, ePackage, mapping editor
Wandora	JVM		-	Java	GPL	github	2023	WebApplication, editor
Django Controlled Vocabularies	Web			Python	BSD	github	2023	eLibrary, CommandLineApplication, editor
Web Protégé	Web		-	Java	BSD	github	2023	WebApplication, WebAPI, SoftwareImage, erverApplication,
Alignment API and Alignment Server	CLI	-		Java	GPL	inria	2023	CommandLineApplication, WebAPI, erverApplication, datab
Neologism	Web			JavaScript	MIT	github	2023	WebApplication, WebAPI, SoftwareImage, erverApplication,
SKOSjs	Web		-	JavaScript	Apache	github	2022	WebApplication, editor
Ginco	Web			Java	GPL	github	2022	WebApplication, WebAPI, erverApplication, editor
VoCol	Web		-	JavaScript	MIT	github	2021	WebApplication, SoftwareImage, erverApplication, editor
mc2skos	CLI		-	Python	Unlicense	github	2021	CommandLineApplication, converter
Skosify	CLI	-		Python	MIT	github	2021	CommandLineApplication, converter, validator
OpenSKOS	Web	-		PHP	GPL	github	2020	WebApplication, WebAPI, SoftwareImage, erverApplication,
SISSVoc	Web	-		XSLT	Apache	github	2019	WebApplication, WebAPI, SoftwareImage, viewer
OntoBee	Web	-		PHP, JavaScript	Apache	github	2018	WebApplication, WebAPI, erverApplication, viewer
SKOS Play	Web	-	-	Java	CC	bitbucket	2018	WebApplication, CommandLineApplication, viewer, converte
SKOS Editor	Web			Java	LGPL	github	2016	WebApplication, WebAPI, erverApplication, editor
HIVE Vocabulary Server	Web		-	Java	BSD	github	2015	WebApplication, erverApplication, viewer
COMA	Web		-	Java	AGPL	sourceforge	2013	DesktopApplication, mapping editor, viewer
ASKOSI	Web	-	-	Java	GPL	askosi	2011	WebApplication, WebAPI, erverApplication, viewer
THManager	JVM		-	Java	LGPL	sourceforge	2006	Application, editor
OntoServer	Web				-	-		WebAPI, erverApplication, viewer
PoolParty Thesaurus Server	Web				-	-		WebApplication, WebAPI, erverApplication, editor
Lexasaurus	Web				-	-		WebApplication, WebAPI, ePackage, erverApplication, edito
Fiblio	Web		-		-	-		WebApplication, erverApplication, ePackage, editor

Please open an issue if some relevant software is missing or data seems invalid!

Metadata

Machine-readable description of the software is based on CodeMeta plus custom fields `api` and `edit`.

Field	Purpose
`name`	name of the software
`url`	Homepage URL
`operatingSystem`	operating system or similar dependency (Web, CLI…)
`api`	whether a web service or similar API is provided to connect to
`edit`	whether vocabularies can be modified with the software
`programmingLanguage`	programming language(s) from SPDX
`license`	license of the software (if free software)
`codeRepository`	source Code repository (if open source)
`dateModified`	year of most recent update
`applicationCategory`	generic type of software
`applicationSubCategory`	type of KOS software

The metadata is also made available in RDF.

KOS software categories

- **viewer**: to display terminologies (23)
- **editor**: to create and modify terminologies (25)
- **mapping editor**: to create and modify mappings/alignments (4)
- **converter**: to convert from one format into another (15)
- **validator**: to check terminologies (11)
- **database**: to store terminologies (3)

Related software

Knowledge organization systems can also be managed with other types software excluded from this report.

More generic software

Simple terminologies can be managed in a **spreadsheet** (LibreOffice Calc, Excel, Google Sheets…). This software lacks most special functionality for terminology management but the usability and accessibility is very high. Some tools in the list above (VocExcel, voc4cat-tool) enhance standard spreadsheets with terminology management functionalities.

The same applies to **database management systems** (RDBMS, NoSQL, RDF triple stores, property graph databases…) with some additional features such as unique key constraints but less usability. An edge case might be systems for management of knowledge graphs (such as Wikibase), included above.

Tools for personal **knowledge management** (such as Obsidian and Notion) help to structure ideas and concepts. Similar tools exist for enterprises to support data integration, knowledge management and/or business intelligence of an organization (for instance PoolParty Semantic Suite).

Several **Content Management Systems** (CMS) allow to manage a list or hierarchy of topics or other entities for knowledge managements. If these lists can be exported, the CMS can be applied to manage terminologies. An example is Drupal, which was used for the first version of BARTOC terminology registry.

Terminologies can also directly be managed in **files** for instance in RDF/Turtle syntax. This only requires a **text editor** but the risk is high to introduce errors. A **version control system** such as git can be used on top to track changes.

More specific software

The term **terminology management** is also used for systems to organize the terms, definitions, and translations used in an organization. See Montoro (2018) for a list of terminology management systems. The focus of these systems is more on translation but there is some overlap with KOS management. Software aimed at simple lists of terms, definitions, glossaries and similar is also excluded from this report.

Several systems and applications exist to manage identifiers (Domain Name System, w3id…). These **identifier systems** are relevant to manage KOS but they are rarely enough for KOS management alone. See Bioregistry for a registry of identifier systems used with KOS.

Several tools, frameworks and programming languages exist for **data transformation** between different formats and/or models (XSLT, jq, Catmandu, DTL, LinkML-map, QVT…). These transformations often imply or make use of terminology mappings.

Some tools for **data curation** also make use of or require terminologies, for instance OpenRefine and mix'n'match. The latter comes close to a mapping editor but it is limited to mappings with Wikidata.

Limited use

Some organizations run **hosted services** for KOS management or related services. Examples include DANTE, xTree and Linked Open Vocabularies (LOV). More examples may be found in the BARTOC list of terminology registries.

programming libraries such as cocoda-sdk, TSS Widgets, jskos-vue, SeMRA, and ng-skos are used to build KOS software.

Other software

The following do not fit into the software categories above or there was not enough information:

- Scripts to analyze concept drift as part of a research project (Open Source)
- TopBraid EDG is a commercial knowledge graph editor
- Apelon DTS (Distributed Terminology System) and Apelon TermManager are commercial terminology editors. An earlier version from 2013 is available as Open Source.
- Grafo is a collaborative knowledge graph editor, available as commercial web application
- Vitro is a framework for semantic web applications, used in VIVO

Related works

This report is based on two reports (Voß 2016a, 2016b) from the beginning of project coli-conc that led to a a wiki page managed between 2020 and 2024. See also Miles and Bechhofer (2009) for a report created during the specification of SKOS W3C Recommendation, the corresponding

wiki pages at https://www.w3.org/2001/sw/wiki/Category:Editor, and Bergman (2018) for a list of software for ontology alignment.

Acknowledgements

Contributions to this report or to its predecessors have been provided by Adrian Pohl, Antoine Isaac, David Linke, Eugene Morozov, Koen Van Daele, Matthias Löbe, Monty Bitto, Roman Baum, Susanne Arndt, and possibly others.

References

Bergman, Michael K. 2018. *30 Active Ontology Alignment Tools*. https://www.mkbergman.com/2129/30-active-ontology-alignment-tools/.

Miles, Alistair, and Sean Bechhofer. 2009. *W3C SKOS Implementation Report*. May. https://www.w3.org/2006/07/SWD/SKOS/reference/20090315/implementation.html.

Montoro, Maria Pia. 2018. *Terminology Management Systems*. https://recremisi.blogspot.com/p/acrolinxterminology-lifecycle.html.

Voß, Jakob. 2016a. *Coli-Conc Technical Report 2: Open Source KOS Software*. March. https://doi.org/10.5281/zenodo.48227.

Voß, Jakob. 2016b. *Open Source Web Applications for Knowledge Organization Systems*. August. https://doi.org/10.5281/zenodo.61262.